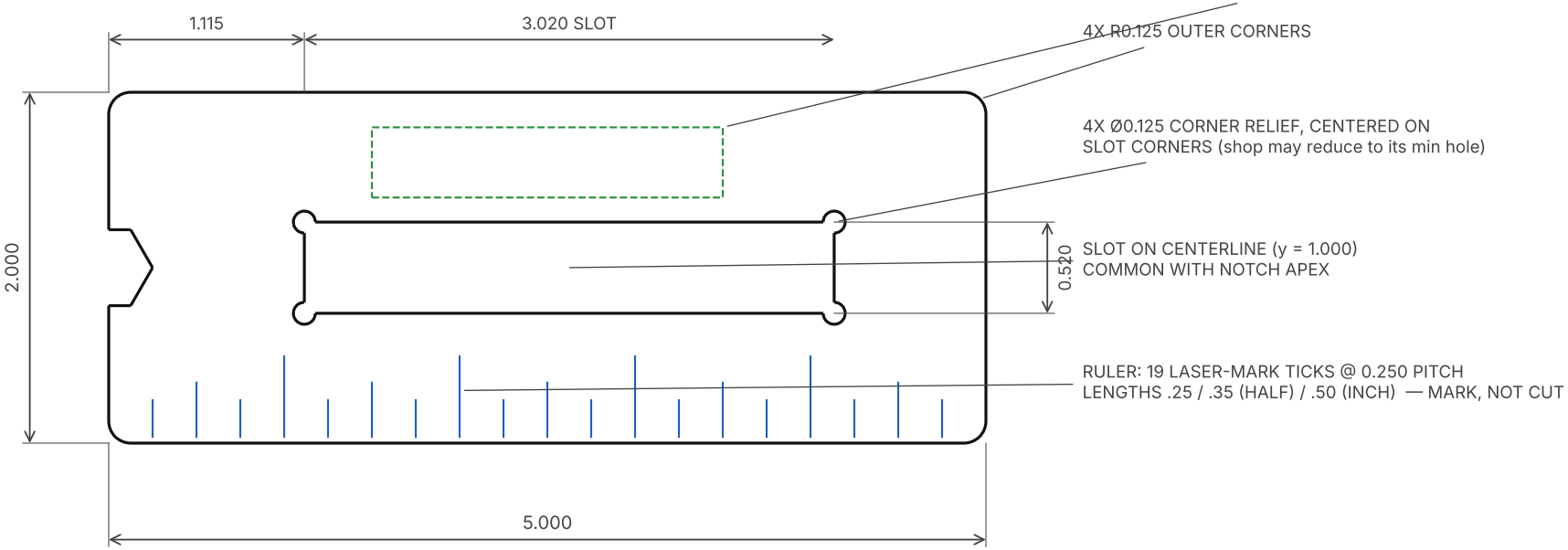
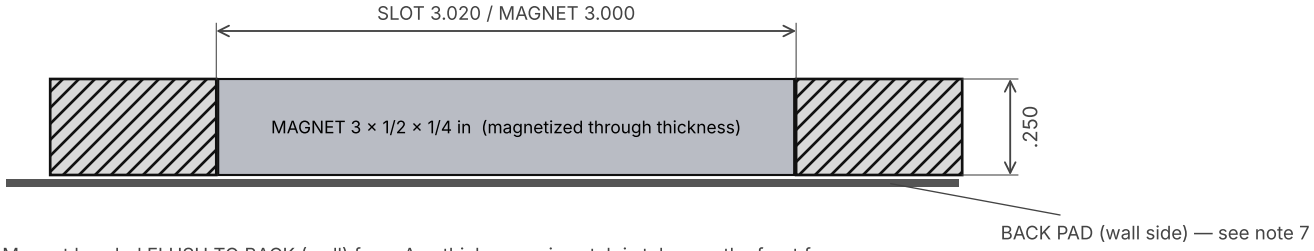
STEADYSTUD v3A — 2D-CUT ALUMINUM BODY, 3 X 1/2 X 1/4 IN MAGNET

Laser/waterjet profile from 0.250" plate. No machining. Dimensions in inches. Drawing generated from the laser DXF.

FRONT FACE (marked side) — 1:1 when PDF printed at 100%

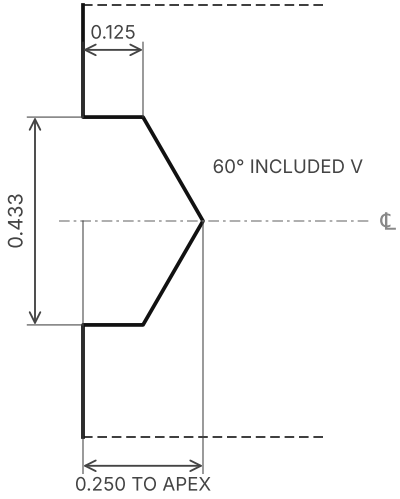


SECTION B — THROUGH CENTERLINE (thickness shown 2:1)



Magnet bonded FLUSH TO BACK (wall) face. Any thickness mismatch is taken on the front face.

DETAIL A — END NOTCH (2.5:1) unchanged from v1 STEP



NOTES

1. MATERIAL: 6061-T6 aluminum plate, 0.250" nominal.
2. PROCESS: through-cut profile only (fiber laser or waterjet). No milling, no chamfers.
3. TOLERANCES: profile ±0.005; slot +0.005/-0.000 vs nominal shown; thickness per mill plate.
4. EDGES: tumble/vibratory deburr. Break all edges .010–.020. No sharp edges (wall-safe).
5. FINISH: Type II anodize, black (alt: clear). Quote as-cut + tumbled as alternate.
6. MARKING: laser mark after anodize — layer MARK (ruler) and LOGO_ZONE. Not engraved.
7. BACK PAD: adhesive-backed PET felt or UHMW tape ≤0.5 mm, die-cut to SteadyStud_v3_back_pad.dxf (outline inset 0.020). Covers magnet; wall-contact face.
8. MAGNET: 3 × 1/2 × 1/4 in NdFeB N42, Ni-Cu-Ni, magnetized through thickness. Source: K&J Magnetics BZ084 (stock). Bond with 2-part epoxy; polarity uniform across lot.
9. USE: hold notch UP (long axis vertical). Mark at notch apex = fastener column = stud line.

PART	SteadyStud body v3A
MAGNET	3 × 1/2 × 1/4 in
SOURCE	cad/v3/steadystud_v3.py
SUPERSEDES	Stud Finer v1 (for quoting)
UNITS	inch
DATE	2026-10-12
STATUS	PROTOTYPE / RFQ — NOT RELEASED